ANALYSTLAB AFRICA

WEEK 8 DATA ANALYTICS CAPSTONE PROJECT

Global Health & Development Dashboard

An Analysis of Health and Development Indicators Using World Bank Data, 2021–2023

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Internship: AnalystLab Africa Data Analytics Internship — Week 8 Capstone Project
Tools: Python, Pandas, Matplotlib/Seaborn, Power BI
Data Source: World Bank World Development Indicators (WDI)

1. Project Title

Global Health & Development: An Analysis of Health and Development Indicators Using World Bank Data, 2021–2023

2. Objective / Problem Statement

Countries differ substantially in health outcomes, economic conditions, healthcare financing, and access to essential infrastructure. Understanding these differences is important for identifying patterns and relationships that can support evidence-based decision-making.

The purpose of this project was to analyse selected World Bank World Development Indicators (WDI) and examine how health outcomes relate to economic conditions, healthcare expenditure, and access to basic sanitation across countries and over the 2021–2023 period.

Objectives

The project aimed to:

- Prepare and clean a large World Bank WDI dataset for analysis.
- Select indicators representing health outcomes, economic conditions, health financing, and sanitation.

Compare countries/economies and examine changes between 2021 and 2023.

Measure relationships among the selected indicators using correlation analysis.

Communicate the findings through an interactive Power BI dashboard.

3. Dataset Description

The project used the **World Bank World Development Indicators (WDI)** dataset.

The original dataset contained **396,970 rows and 70 columns**, with annual observations covering **1960–2025** and **1,498 unique indicators**.

Because the dataset was very large, the analysis was narrowed to five indicators that were considered most relevant to the project's health and development objectives.

Selected Indicators

Indicator	Purpose
Life expectancy at birth, total (years)	Population-health outcome
Mortality rate, infant (per 1,000 live births)	Infant and population health outcome
GDP per capita (constant 2015 US$)	Economic development indicator
Current health expenditure (% of GDP)	Health financing indicator
People using at least basic sanitation services (% of population)	Essential sanitation/infrastructure indicator

The analysis focused primarily on the period **2021–2023** because this period provided comparatively stronger and more consistent coverage for the selected indicators.

4. Data Cleaning Process

The data-cleaning process was carried out using **Python and Pandas**.

The main steps included:

Inspecting the dataset dimensions, columns, data types, and missing values.

Reviewing the available indicators and identifying those relevant to the project.

Assessing indicator coverage for 2021–2023.

Selecting five indicators with suitable coverage.

Excluding the **Physicians (per 1,000 people)** indicator because its coverage was substantially lower and inconsistent, particularly in 2023.

Removing five **"Not classified"** records that had no usable observations.

Checking for duplicate records.

Checking indicator ranges for implausible values.

Retaining unavailable observations as missing values rather than applying arbitrary imputation.

Creating and verifying the final analysis-ready dataset.

Indicator Coverage

Indicator	2021	2022	2023
Current health expenditure (% GDP)	90.57%	90.19%	90.19%
GDP per capita	96.23%	96.23%	94.34%
Life expectancy	99.62%	99.62%	99.62%
Infant mortality	91.70%	91.70%	91.70%
Basic sanitation	93.58%	92.83%	91.70%
Physicians per 1,000 people	46.42%	64.53%	21.89%

The coverage assessment demonstrated why the Physicians indicator was excluded from the final analysis.

The final complete-case **2023 dataset contained 221 rows and 7 columns**, with **zero missing values** in the selected fields.

5. Analysis Methodology

The project followed an end-to-end data analytics workflow consisting of:

Data Exploration

The original WDI dataset was explored to understand its structure, size, indicators, years, missing values, and data coverage.

Data Cleaning

The dataset was filtered and transformed using Pandas. Indicators with inadequate coverage were removed and the final dataset was validated for duplicates, missing values, and implausible values.

Descriptive Analysis

Descriptive statistics were calculated to understand the central tendency and range of the selected indicators.

Trend Analysis

Average values were compared across 2021, 2022, and 2023 to identify changes over time.

Country Comparison

Countries/economies were compared using the selected 2023 indicators to highlight differences in health and development outcomes.

Correlation Analysis

Pearson correlation coefficients were calculated using the 2023 complete-case dataset to examine relationships between the selected indicators.

Data Visualization

Python was used for exploratory visualizations, while **Power BI** was used to create the final interactive dashboard.

> **Important:** Correlation indicates an association between variables. It does not prove that one variable causes another.

6. Key Findings

The analysis identified several changes between 2021 and 2023.

Indicator	2021 Average	2023 Average	Change
Life expectancy	71.47 years	73.49 years	+2.83%
Infant mortality	21.18	20.22	−4.54%
GDP per capita	US$16,068.51	US$16,898.66	+5.17%
Basic sanitation	78.33%	79.66%	+1.69%
Health expenditure	7.06%	6.73%	−4.59%

Main findings

Average **life expectancy increased** from 71.47 years in 2021 to 73.49 years in 2023.

Average **GDP per capita increased by approximately 5.17%**.

Average access to **basic sanitation increased** from 78.33% to 79.66%.

Average **infant mortality decreased** from 21.18 to 20.22 deaths per 1,000 live births.

Average **current health expenditure as a percentage of GDP decreased** from 7.06% to 6.73%.

7. Insights

The 2023 correlation analysis identified several important relationships.

Relationship	Pearson r	Interpretation
Life expectancy ↔ Infant mortality	−0.899	Strong negative association
Life expectancy ↔ Basic sanitation	+0.834	Strong positive association
Infant mortality ↔ Basic sanitation	−0.859	Strong negative association
Life expectancy ↔ GDP per capita	+0.631	Moderate positive association
Infant mortality ↔ GDP per capita	−0.484	Moderate negative association

Sanitation and Health

One of the strongest findings was the relationship between sanitation coverage and infant mortality.

The correlation was:

$r = -0.859$

This indicates a strong negative association. In the 2023 data, countries with higher levels of basic sanitation coverage generally tended to have lower infant mortality.

Sanitation also showed a strong positive relationship with life expectancy:

r = +0.834

This means countries with higher sanitation coverage generally tended to have higher life expectancy.

GDP and Life Expectancy

GDP per capita showed a moderate positive relationship with life expectancy:

r = +0.631

This suggests that countries with higher GDP per capita generally tended to have higher life expectancy.

GDP per capita also had a moderate negative relationship with infant mortality:

r = −0.484

Overall, the findings suggest that health outcomes are associated with broader socioeconomic and infrastructure conditions.

However, these relationships **should not be interpreted as proof of causation**.

8. Power BI Dashboard

The analysis was converted into an interactive Power BI dashboard titled:

Global Health & Development Dashboard

World Bank Indicators | 2021–2023

The dashboard contains:

- Five KPI cards:
    - Life expectancy
    - Infant mortality
    - GDP per capita
    - Basic sanitation
    - Health expenditure
- Year slicer
- Country Name slicer

Indicator Name slicer

Life expectancy trend chart

GDP per capita trend chart

Basic sanitation trend chart

Sanitation coverage vs infant mortality scatter plot

GDP per capita vs life expectancy scatter plot

Key Findings section

Dashboard Screenshot

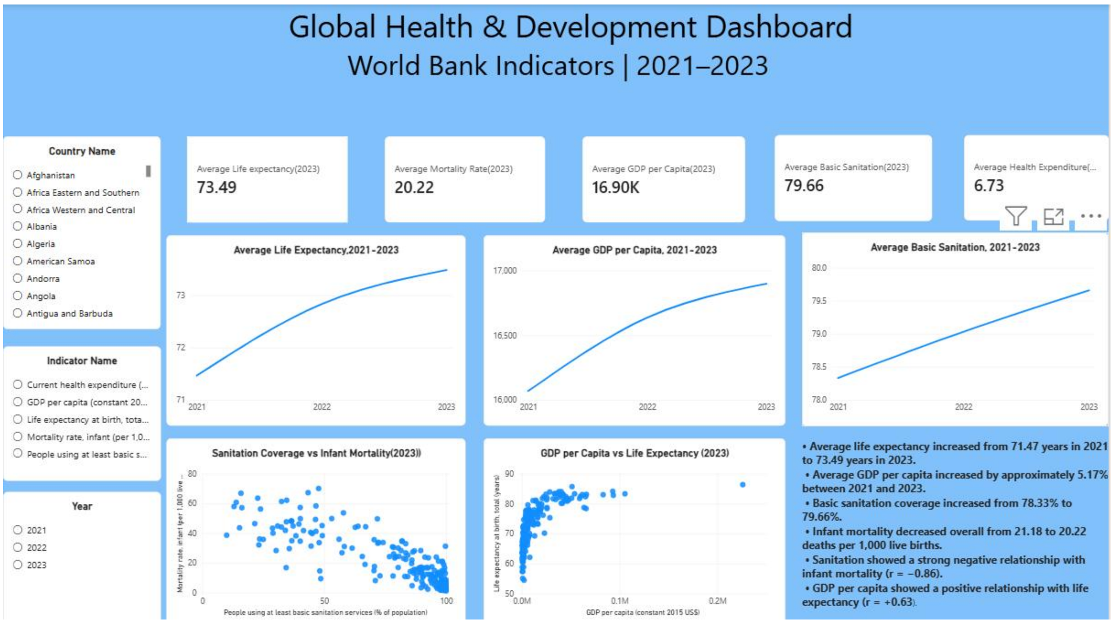

Figure 1: Final Global Health & Development Power BI Dashboard.

The dashboard allows users to interactively explore the data by year, country, and indicator while also viewing overall trends and relationships.

9. Recommendations

Based on the analysis, the following recommendations are proposed:

1. Strengthen access to basic sanitation

Greater attention should be given to sanitation infrastructure, particularly in countries where low sanitation coverage coincides with poor health outcomes.

2. Consider broader development conditions

Health outcomes should be assessed alongside economic and infrastructure conditions rather than focusing on healthcare expenditure alone.

3. Use multiple indicators

Decision-makers should consider several indicators together when identifying countries or economies facing development and health challenges.

4. Improve data quality and coverage

More consistent and complete health and development data collection would improve the reliability of international comparisons and trend analysis.

5. Promote data-driven decision-making

Interactive dashboards such as the Power BI dashboard developed in this project can make large datasets easier for decision-makers and stakeholders to understand.

6. Investigate relationships further

The correlations identified in this project provide useful starting points for deeper research. Future studies could investigate causal relationships using longer time periods and additional variables.

10. Conclusion

This project demonstrates how a large real-world World Bank dataset can be transformed into an analysis-ready dataset and ultimately into an interactive data visualization dashboard.

The analysis identified substantial differences in health outcomes, economic conditions, healthcare expenditure, and sanitation access across countries and economies.

Between 2021 and 2023, average **life expectancy, GDP per capita, and basic sanitation access increased**, while **infant mortality and health expenditure as a percentage of GDP decreased**.

The 2023 correlation analysis highlighted particularly strong relationships between sanitation and health outcomes. Higher sanitation coverage was associated with higher life expectancy and lower infant mortality. GDP per capita also showed a moderate positive association with life expectancy.

Overall, the findings suggest that population health is connected to wider economic and infrastructure conditions. The project also demonstrates the importance of **data-**

quality assessment, appropriate indicator selection, statistical analysis, and clear visual communication when working with large datasets.

The completed Power BI dashboard provides an accessible way to communicate these findings and allows users to explore the indicators interactively.

11. Tools and Data Source

Tools Used

Python and Pandas
Used for data loading, cleaning, filtering, transformation, validation, descriptive analysis, and correlation analysis.

Matplotlib and Seaborn
Used for exploratory data visualization during the analytical stage.

Power BI
Used to create the final interactive dashboard and communicate the analytical findings.

Data Source

World Bank World Development Indicators (WDI)

The WDI dataset provides internationally comparable economic, social, health, and development indicators for countries and economies around the world.

12. Limitations

The project has several limitations:

- The analysis covers **2021–2023**, which is a relatively short period for assessing long-term development trends.

- Data coverage differs between indicators and countries.

- The Physicians indicator was excluded because of inconsistent coverage, particularly in 2023.

- The WDI dataset contains country/economy records and may include aggregate entities.

- Correlation analysis identifies association and does not establish causation.

- The five selected indicators do not capture every social, economic, environmental, or health-system factor that may influence health outcomes.